

RetroDiff: Training Objective Derivation

Diffusion models are trained by maximising a variational lower bound on the log-likelihood. Applying Jensen’s inequality to the intractable marginal $\log p_\theta(x)$ gives the *evidence lower bound* (ELBO):

$$\log p_\theta(x) \geq \mathbb{E}_{q(x_{1:T}|x)} \left[\log \frac{p_\theta(x_{0:T})}{q(x_{1:T}|x)} \right] = -\mathcal{L}_{\text{ELBO}} \quad (1)$$

where q is the fixed forward noising process, p_θ is the learned reverse process parameterised by θ , and $x_{1:T}$ denotes the latent variables across all T diffusion steps.

Expanding the joint ratio in (1) via the Markov structure of both p_θ and q yields a telescoping decomposition into a reconstruction term and a sum of per-step KL divergences:

$$\mathcal{L}_{\text{ELBO}} = \underbrace{\mathbb{E}_q[-\log p_\theta(x_0|x_1)]}_{\text{reconstruction}} + \sum_{t=2}^T \underbrace{\text{KL}(q(x_{t-1}|x_t, x) \parallel p_\theta(x_{t-1}|x_t))}_{\text{step-}t \text{ denoising}} \quad (2)$$

Each KL term in (2) can be computed in closed form because the forward posterior $q(x_{t-1}|x_t, x)$ is Gaussian; minimising these terms trains the network to reverse each noising step.

RetroDiff’s contribution. Standard diffusion models generate x from noise alone. RetroDiff conditions every denoising step on a set $R(x) = \{r_1, \dots, r_k\}$ of k exemplars retrieved from the training corpus based on x . Substituting $\mathcal{L}_{\text{ELBO}}(x | R(x))$ for $\mathcal{L}_{\text{ELBO}}(x)$ in the per-sample objective and taking an expectation over the data distribution defines the full training loss:

$$\mathcal{L}_{\text{RetroDiff}}(\theta) = \mathbb{E}_{x \sim p_{\text{data}}} [\mathcal{L}_{\text{ELBO}}(x | R(x))] \quad (3)$$

Equation (3) is the central objective: the model must learn to denoise x given both the noisy intermediate x_t and the retrieved exemplar context, making retrieval a first-class component of the generative process rather than a post-hoc conditioning signal.

Symbol	Meaning
x	clean data sample
x_t	noised version of x at diffusion step t
$R(x)$	set of k retrieved exemplars conditioned on x
θ	model parameters
q	forward noising distribution
p_θ	learned reverse distribution
T	total diffusion steps